Skipper-CCD background rejection

UBA - Gravitational Waves and AI-Assisted Research

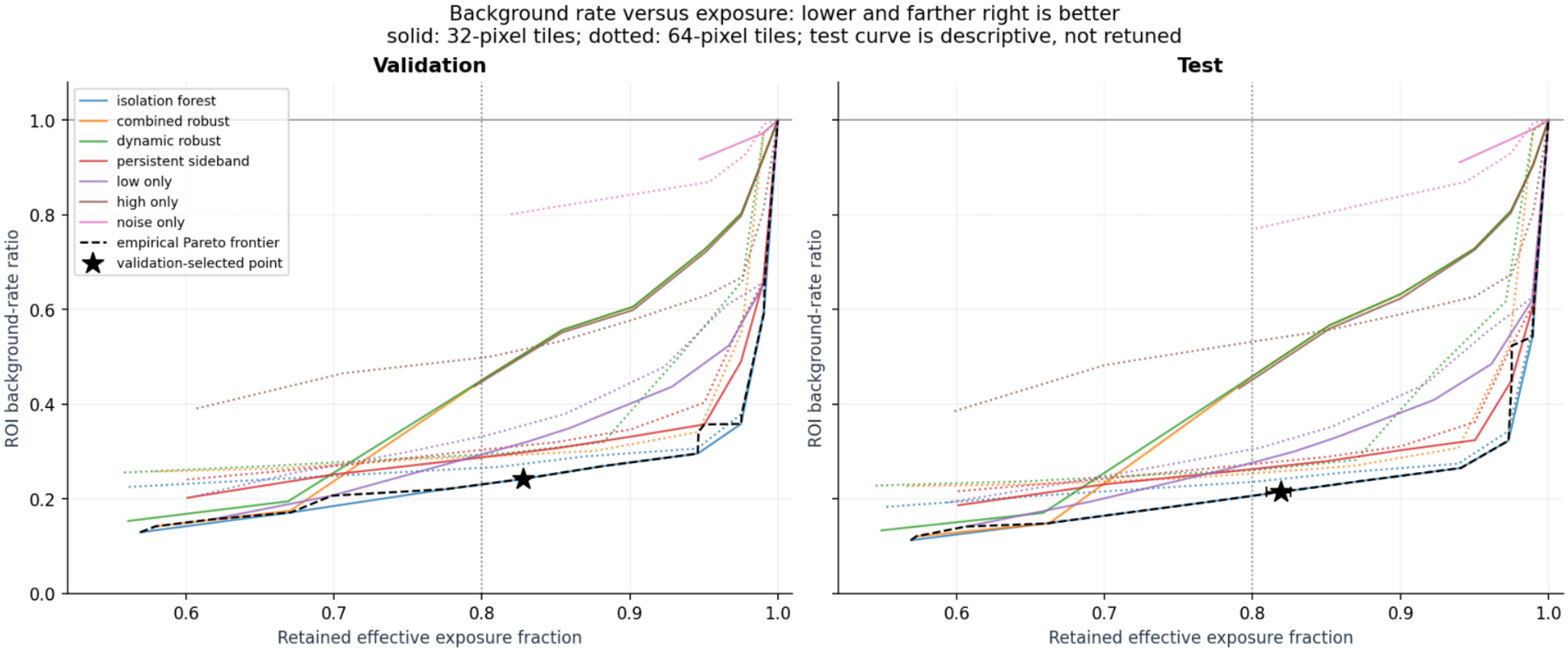
Held-out result: 81.91% of exposure retained; ROI rate reduced by 78.57%. Counts: 22,237 to 3,903. Rate: 812.43 to 174.09 per million pixel-days. F_equal = 3.822.

Method: 32 x 32 pixel patches; six ROI-blind sideband/persistence/noise descriptors; unsupervised Isolation Forest (100 trees, 256 samples/tree). Whole regions are masked; events are evaluated by their reconstructed maximum-charge anchor.

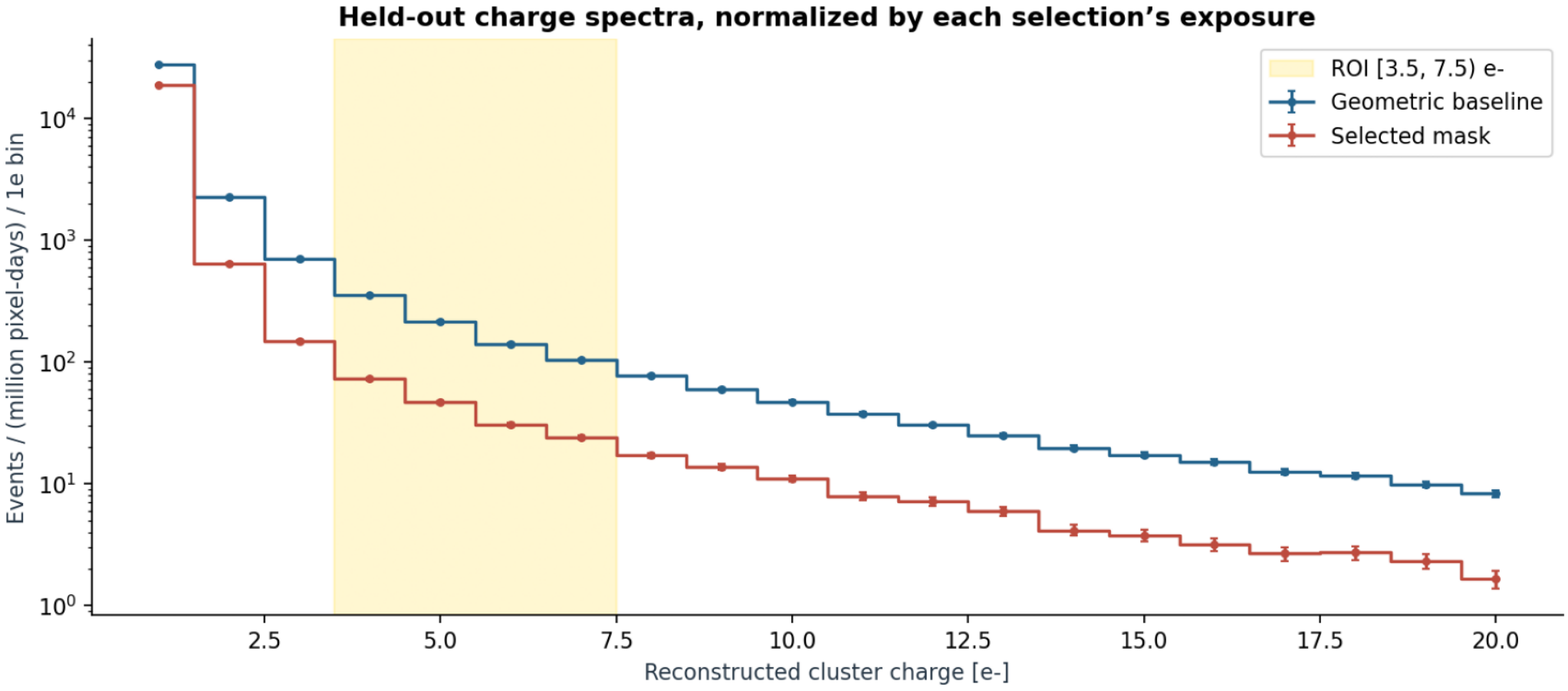
Validation: 65 training acquisitions, 22 validation, 22 chronological held-out test; both modules and all HDUs stay together. Selection maximizes validation F_equal subject to at least 80% global exposure. Test is not retuned.

Limitations: gain is concentrated in S101 HDU0. Baseline is active geometry, not production masks. Exposure assumes uniform readout timing; no kg-day claim or measured true-signal efficiency. Bootstrap intervals are conditional. Same-agent numerical verification, no independent scientist review.

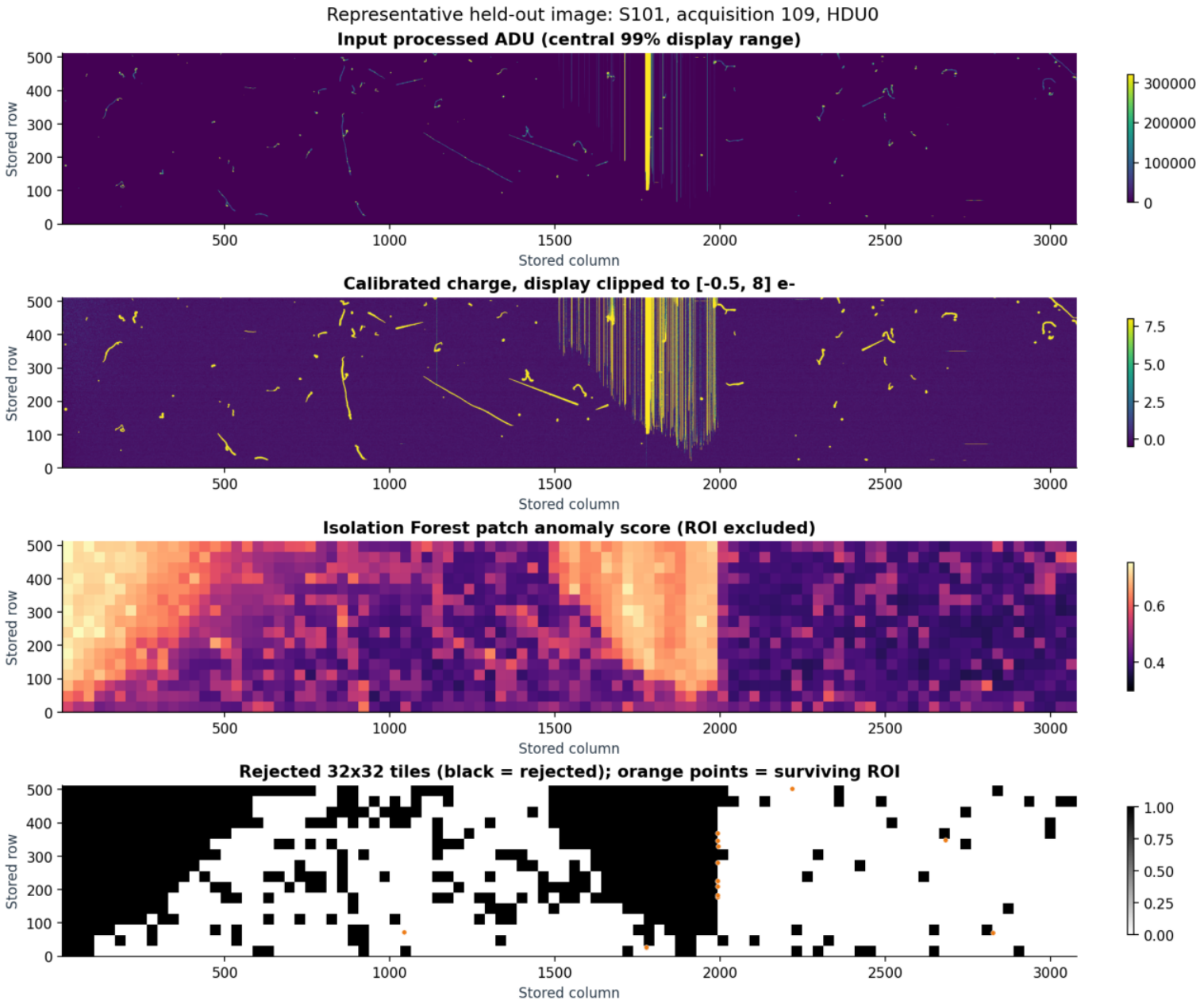
Reproduce: SKIPPER_DATA_DIR=/path/to/pp_proc bash CODEX/gpt-6-astra_high/reproduce.sh. Read report.html, METHOD.md and PROVENANCE.md for full definitions, sources and checks. No FITS data are stored in Git.



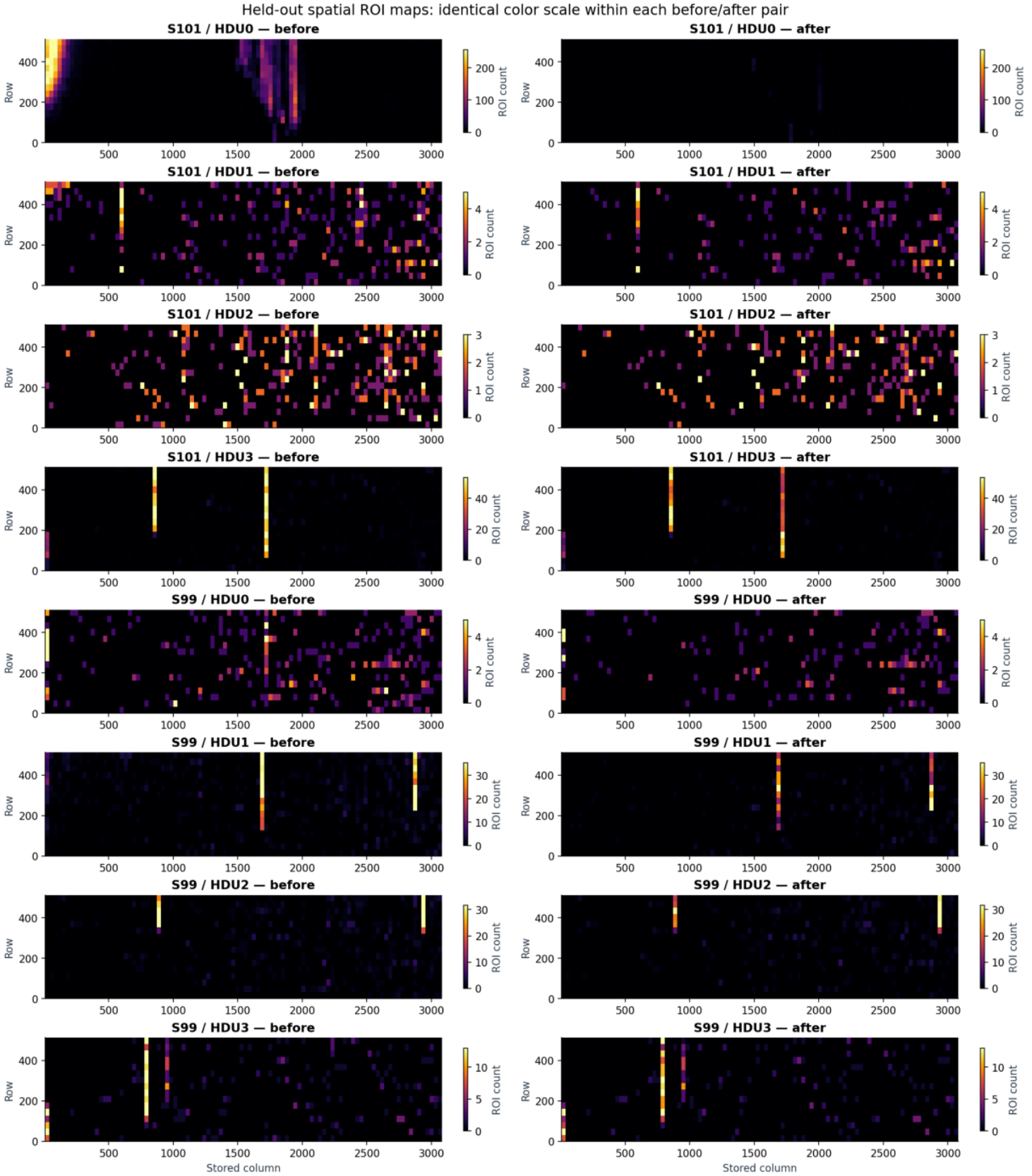
Every tested operating point; frontier includes comparison methods. The star is fixed by validation. Error bars: paired four-acquisition block bootstrap, 95% intervals, conditional on the model. Lower rate and greater exposure are preferred.



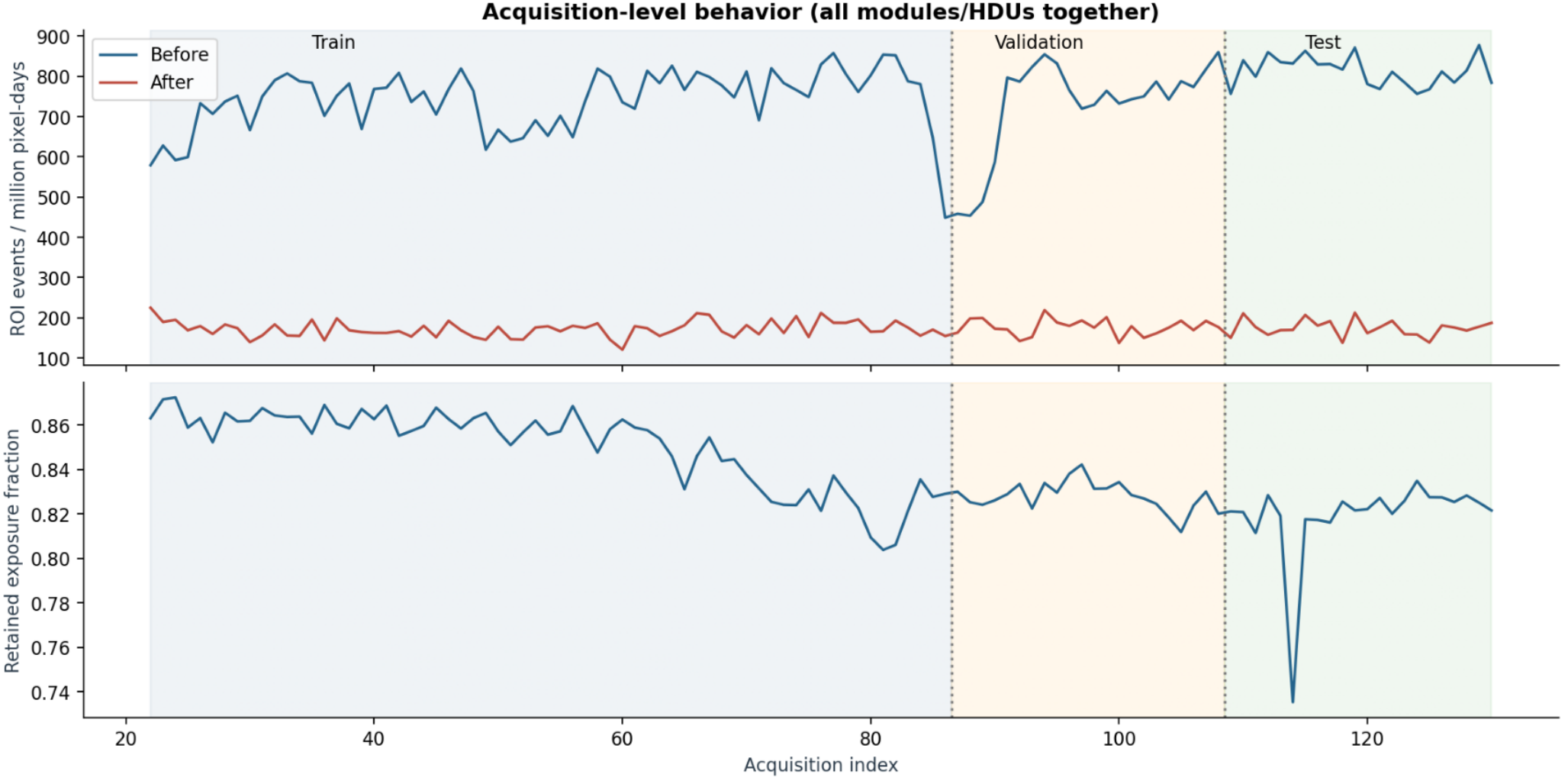
Clusters are reconstructed before masking; selection uses the anchor tile. Each spectrum is divided by its own surviving pixel-days. Illustrative Poisson errors do not include calibration or temporal systematics.



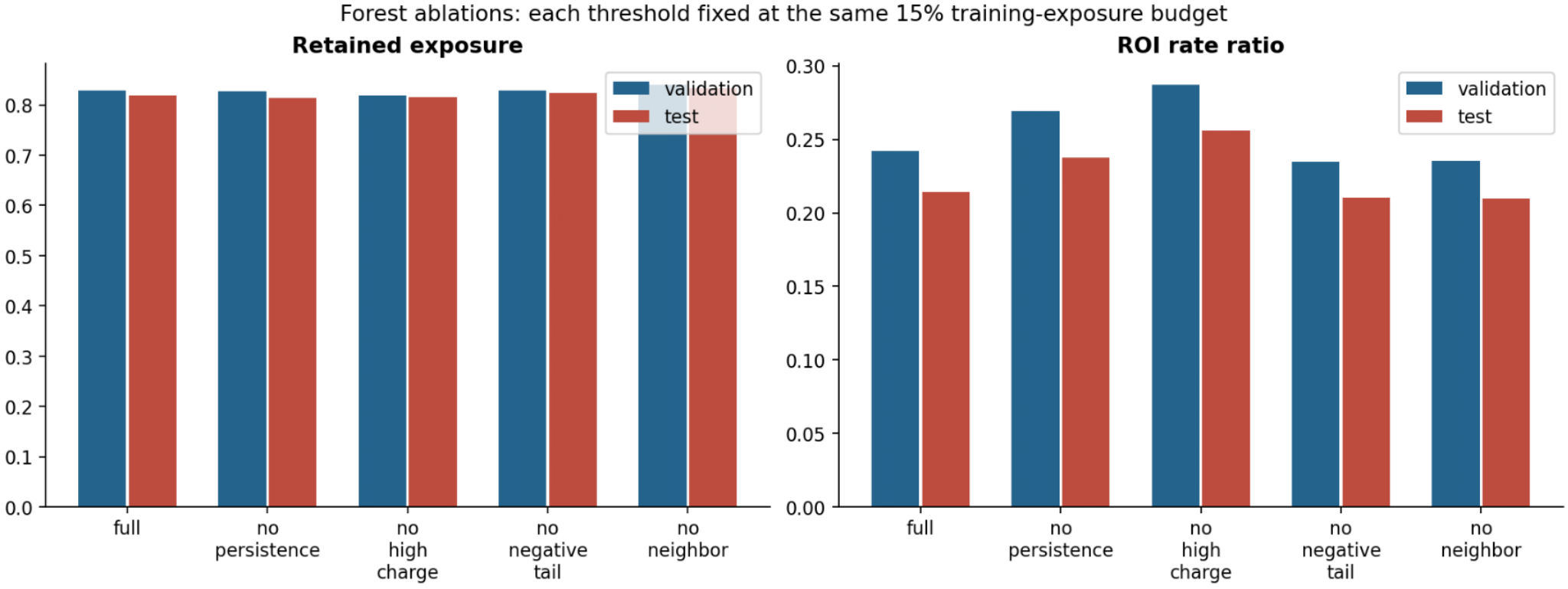
First test acquisition, chosen by index, not by best visual performance. Coordinates are stored FITS coordinates. The score has no direct ROI component input. Display clipping does not alter numerical calculations.



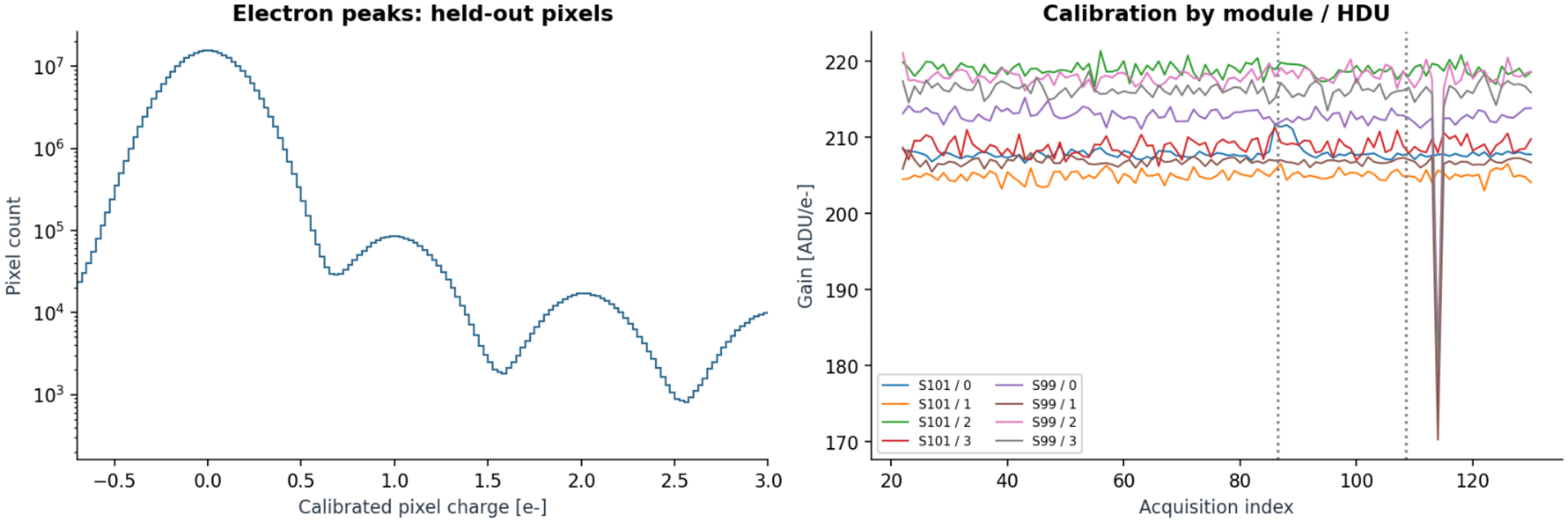
All eight module/HDU combinations. These are counts per 32x32 tile, not exposure-corrected rates. Each pair shares a scale clipped at the baseline 99.5th percentile for visibility; exposure-normalized results appear in the adjacent table.



No random pixel split: paired modules and four HDUs each stay in one acquisition block. Training performance is in-sample; only the final green block is held out.



Removing one component from the selected Forest descriptor, with refitting on training only. These checks were not used to retune the final method; lower rate at lower exposure is not automatically better.



Electron peaks and gain stability. Vertical separators are the frozen acquisition boundaries. The baseline is active geometry only, without MOSKITA production masks.